

index.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/index.py (3.8.3)

File Edit Format Run Options Window Help

```
#Commencement of python programming

print("Hello World")
name=input("Enter your name: ")
print("My name is",name)
print("Hello "+name+" ,Welcome to the python class")
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/index.py =====
Hello World
Enter your name: Jaideep
My name is Jaideep
Hello Jaideep ,Welcome to the python class
>>> |
```

Area_circle.pyw - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Area_circle.pyw (3.8.3)

File Edit Format Run Options Window Help

#Write a Python program which accepts the radius of a circle from the user and compute the area.

#AREA OF CIRCLE

```
radius=int(input("Enter radius: "))
area=3.14*(radius**2)
print(f"Area of a circle is {area}")      #This is f-string
#print(f"Area of a circle: {3.14*(radius**2)}")
```

Python 3.8.3 Shell

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Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32

Type "help", "copyright", "credits" or "license()" for more information.

>>>

==== RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Area_circle.pyw ===

Enter radius: 5

Area of a circle is 78.5

>>>

Largest Number.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Largest Number.py (3.8.3)

File Edit Format Run Options Window Help

#Write a Python program to get the largest number from a list.

```
numbers=[10,45,36,89,12]
print("Index:",numbers[-1])
print("Largest Number from the array:",max(numbers))
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
=== RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Largest Number.py ===
Index: 12
Largest Number from the array: 89
>>> |
```

Color.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Color.py (3.8.3)

File Edit Format Run Options Window Help

#Write a Python program to display the first and last colors from the following list.

```
color_list = ["Red","Green","White" ,"Black"]  
print("First color:",color_list[0])  
print("Last color:",color_list[-1])
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Color.py =====
First color: Red
Last color: Black
>>> |

Sum or thrice.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Sum or thrice.py (3.8.3)

File Edit Format Run Options Window Help

```
'''Write a Python program to calculate the sum of three given numbers,  
if the values are equal then return thrice of their sum.'''
```

```
#Calculate sum of three numbers
```

```
a=int(input("Enter 1st number: "))
```

```
b=int(input("Enter 2nd number: "))
```

```
c=int(input("Enter 3rd number: "))
```

```
sum=(a+b+c)
```

```
if(a==b==c):
```

```
    print ("Result: ",sum*3)
```

```
else:
```

```
    print ("Result: ",sum)
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32

Type "help", "copyright", "credits" or "license()" for more information.

>>>

=== RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Sum or thrice.py ===

Enter 1st number: 95

Enter 2nd number: 96

Enter 3rd number: 97

Result: 288

>>>

=== RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Sum or thrice.py ===

Enter 1st number: 95

Enter 2nd number: 95

Enter 3rd number: 95

Result: 855

>>>

Mini_Project.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Mini_Project.py (3.8.3)

File Edit Format Run Options Window Help

```
#Build a simple grade calculator
```

```
math=int(input("Enter your maths marks: "))
eng=int(input("Enter your english marks: "))
evs=int(input("Enter your evs marks: "))
ssc=int(input("Enter your ssc marks: "))
hindi=int(input("Enter your hindi marks: "))
total_marks=math+eng+evs+ssc+hindi
perc=(total_marks)/5
if(perc>=90):
    print("Your grade: A")
elif(perc>=80):
    print("Your grade: B")
elif(perc>=70):
    print("Your grade: C")
elif(perc>=60):
    print("Your grade: D")
elif(perc>=50):
    print("Your grade: E")
else:
    print("Fail")
print(f"Your Total marks: {total_marks}")
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
==== RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Mini_Project.py ===
Enter your maths marks: 95
Enter your english marks: 81
Enter your evs marks: 75
Enter your ssc marks: 45
Enter your hindi marks: 55
Your grade: C
Your Total marks: 351
>>> |

Even or Odd.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Even or Odd.py (3.8.3)

File Edit Format Run Options Window Help

```
'''Write a Python program to find whether a given number (accept from the user)
    is even or odd, print out an appropriate message to the user.'''
```

```
#Display number is even or odd
```

```
d=int(input("Enter number: "))
if(d%2==0):
    print("Number",d,"is even")
else:
    print("Number",d,"is odd")
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32

Type "help", "copyright", "credits" or "license()" for more information.

>>>

==== RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Even or Odd.py ====

Enter number: 5

Number 5 is odd

>>>

==== RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Even or Odd.py ====

Enter number: 10

Number 10 is even

>>> |

Sum of all numbers.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Sum of all numbers.py (3.8.3)

File Edit Format Run Options Window Help

```
#Write a Python function to sum all the numbers in a list.
```

```
#To print sum of all numbers in a list using function
```

```
def list_sum(list):
```

```
    sum=0
```

```
    for i in list:
```

```
        sum+=i
```

```
    return sum
```

```
numbers=[1,2,3,4,5]
```

```
print("Sum of all items which is present in the list is",list_sum(numbers))
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32

Type "help", "copyright", "credits" or "license()" for more information.

>>>

= RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Sum of all numbers.py

Sum of all items which is present in the list is 15

>>> |

Add a key to dictionary.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Add a key to dictionary.py (3.8.3)

File Edit Format Run Options Window Help

#Write a Python script to add a key to a dictionary.

#Program to add a key to dictionary

```
my_dict={1:"Apple",2:"Banana",3:"Mango"}
print("Original dictionary:",my_dict)
my_dict[4]="Orange"
print("Updated dictionary:",my_dict)
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32

Type "help", "copyright", "credits" or "license()" for more information.

>>>

= RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Add a key to dictionary.py

Original dictionary: {1: 'Apple', 2: 'Banana', 3: 'Mango'}

Updated dictionary: {1: 'Apple', 2: 'Banana', 3: 'Mango', 4: 'Orange'}

>>> |

Check for a given key.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Check for a given key.py (3.8.3)

File Edit Format Run Options Window Help

#Write a Python script to check if a given key already exists in a dictionary.

#Program to check if a given key already exists in a dictionary

```
my_dict={1:"Apple",2:"Banana",3:"Mango"}
key=int(input("Enter key to check: "))
if key in my_dict:
    print("Key exists")
else:
    print("Key doesn't exists")
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Check for a given key.py
Enter key to check: 3
Key exists
>>>
= RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Check for a given key.py
Enter key to check: 5
Key doesn't exists
>>> |
```

For learning.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/For learning.py (3.8.3)

File Edit Format Run Options Window Help

```
#For learning
```

```
list1=[1,2,3]
list2=[4,5,"RAHUL"]
list3=list1+list2
print(list3)
list4=[7,8,9]
list3.extend(list4)
print(list3)
list3.insert(9,10)
print(list3)
list3.append(11)
print(list3)
del list3[10]
print(list3)
del list3[1:4]
print(list3)
list3.remove(8)
print(list3)
list3.pop()
print(list3)
list3.clear()
print(list3)
my_profiles={'Name':'Rahul','Age':20,'E-mail':'abc@gmail.com','Contact no.':9810504681}
print(my_profiles)
print(my_profiles['Age'])
my_profiles['City']='New York'
print(my_profiles)
del my_profiles["Age"]
print(my_profiles)
print(my_profiles.keys())
print(my_profiles.values())
```

```
Python 3.8.3 Shell
File Edit Shell Debug Options Window Help
Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
==== RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/For learning.py ====
[1, 2, 3, 4, 5, 'RAHUL']
[1, 2, 3, 4, 5, 'RAHUL', 7, 8, 9]
[1, 2, 3, 4, 5, 'RAHUL', 7, 8, 9, 10]
[1, 2, 3, 4, 5, 'RAHUL', 7, 8, 9, 10, 11]
[1, 2, 3, 4, 5, 'RAHUL', 7, 8, 9, 10]
[1, 5, 'RAHUL', 7, 8, 9, 10]
[1, 5, 'RAHUL', 7, 9, 10]
[1, 5, 'RAHUL', 7, 9]
[]
{'Name': 'Rahul', 'Age': 20, 'E-mail': 'abc@gmail.com', 'Contact no.': 9810504681}
20
{'Name': 'Rahul', 'Age': 20, 'E-mail': 'abc@gmail.com', 'Contact no.': 9810504681, 'City': 'New York'}
{'Name': 'Rahul', 'E-mail': 'abc@gmail.com', 'Contact no.': 9810504681, 'City': 'New York'}
dict_keys(['Name', 'E-mail', 'Contact no.', 'City'])
dict_values(['Rahul', 'abc@gmail.com', 9810504681, 'New York'])
>>> [
```

File_Handling_1.py - C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python\File_Handling_1.py (3.8.3)

File Edit Format Run Options Window Help

#FILE HANDLING 1

#Writing to a file

```
f=open("sample.txt","w")
f.write("Hello, this is a file handling example in Python.")
f.close()
```

#Reading from a file

```
f=open("sample.txt","r")
print("File Content: ")
print (f.read())
f.close()
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32

Type "help", "copyright", "credits" or "license()" for more information.

>>>

== RESTART: C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python\File_Handling_1.py ==

File Content:

Hello, this is a file handling example in Python.

>>> |



File_Handling_2.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/File_Handling_2.py (3.8.3)

File Edit Format Run Options Window Help

#FILE HANDLING 2

```
lines=["First line\n", "Second line\n"]
with open("new_file.txt", "w") as file:
    file.write("Hello World!\n")
    file.writelines(lines)
```

File_Handling_3.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/File_Handling_3.py (3.8.3)

File Edit Format Run Options Window Help

#FILE HANDLING 3

```
import csv
data=[{"Name":"Alice", "Score":85}, {"Name": "Bob", "Score":92}, {"Name": "Charlie", "Score":78}]

#Writing to csv

with open("grades.csv", "w", newline="") as f:
    writer=csv.DictWriter(f, fieldnames=["Name", "Score"])
    writer.writeheader()
    writer.writerows(data)

#Reading it back

with open("grades.csv", "r", newline="") as f:
    reader=csv.DictReader(f)
    for row in reader:
        print(f"{row['Name']} scored {row['Score']}")
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32

Type "help", "copyright", "credits" or "license()" for more information.

>>>

== RESTART: C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python\File_Handling_3.py ==

Alice scored 85

Bob scored 92

Charlie scored 78

>>>

Desktop

File Working Directory.py - C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python\File Working Directory.py (3.8.3)

File Edit Format Run Options Window Help

#To print file present working directory.

```
import os
filewd=os.getcwd()
print(filewd)
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32

Type "help", "copyright", "credits" or "license()" for more information.

>>>

= RESTART: C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python\File Working Directory .PY

C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python

>>> |



Exception_Handling.py - C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python\Exception_Handling.py (3.8.3)

File Edit Format Run Options Window Help

#Exception Handling

```
try:
    number=int(input("Enter your integer number to divide by 10: "))
    result=10/number
    print(f"Result is {result}")
except ZeroDivisionError:
    #Runs if you are trying to divide by zero
    print("Error: Can't divide by zero")
except ValueError:
    #Runs if passes string value to number
    print("Error: Enter a valid number")
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1916 64-bit] on win32

Type "help", "copyright", "credits" or "license()" for more information

>>>

= RESTART: C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python\Exception_Handling.py

Enter your integer number to divide by 10: 0

Error: Can't divide by zero

>>>

= RESTART: C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python\Exception_Handling.py

Enter your integer number to divide by 10: -5

Result is -2.0

>>>

= RESTART: C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python\Exception_Handling.py

Enter your integer number to divide by 10: jai

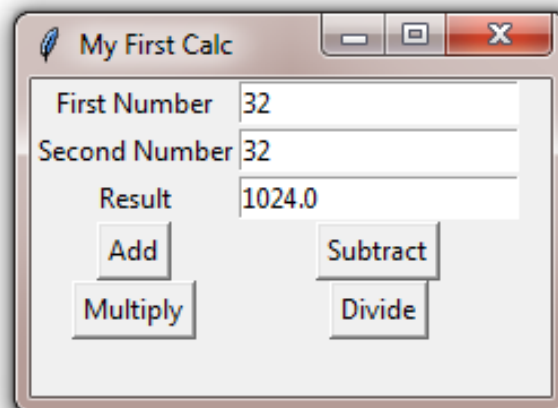
Error: Enter a valid number



File Edit Format Run Options Window Help

#Write a Python script to make calculator using Tkinter.

```
import tkinter as tk
root=tk.Tk()
def add():
    result.set(float(e1.get())+float(e2.get()))
def sub():
    result.set(float(e1.get())-float(e2.get()))
def mul():
    result.set(float(e1.get())*float(e2.get()))
def div():
    result.set(float(e1.get())/float(e2.get()))
root.title("My First Calc")
root.geometry("230x140")
tk.Label(root,text="First Number").grid(row=0,column=0)
tk.Label(root,text="Second Number").grid(row=1,column=0)
e1=tk.Entry(root)
e2=tk.Entry(root)
e1.grid(row=0,column=1)
e2.grid(row=1,column=1)
result=tk.StringVar()
tk.Label(root,text="Result").grid(row=2,column=0)
tk.Entry(root,textvariable=result).grid(row=2,column=1)
tk.Button(root,text="Add",command=add).grid(row=3,column=0)
tk.Button(root,text="Subtract",command=sub).grid(row=3,column=1)
tk.Button(root,text="Multiply",command=mul).grid(row=4,column=0)
tk.Button(root,text="Divide",command=div).grid(row=4,column=1)
root.mainloop()
```



Stack Implementation using list.py - C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python\Stack Implementatio
File Edit Format Run Options Window Help

#Stack implementation using list

```
stack=[]
```

```
#Elements push onto stack(PUSH operation)
```

```
stack.append(10)
```

```
stack.append(20)
```

```
stack.append(30)
```

```
print("Stack after push elements:",stack)
```

```
#Elements pop from last(POP operation)
```

```
if stack:
```

```
    removed=stack.pop()
```

```
    print("Stack element popped:",removed)
```

```
    print("Stack after pop:",stack)
```

```
else:
```

```
    print("Stack is empty!")
```

```
#Access top element from stack(Peek element)
```

```
if stack:
```

```
    print("Top element:",stack[-1])
```

```
else:
```

```
    print("Stack is empty!")
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python\Stack Implementation using list.py
Stack after push elements: [10, 20, 30]
Stack element popped: 30
Stack after pop: [10, 20]
Top element: 20
>>> |

Queue Implementation using list.py - C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python\Queue Implementation using list

File Edit Format Run Options Window Help

#Queue Implementation using list

```
queue=[]
```

#Queue enqueue operation

```
queue.append(10)
```

```
queue.append(20)
```

```
queue.append(30)
```

```
print("Queue after enqueue operation:",queue)
```

#Queue dequeue operation

```
if queue:
```

```
    dequeue=queue.pop(0)
```

```
    print("Dequeued element from queue:",dequeue)
```

```
    print("Queue after dequeue operation:",queue)
```

```
else:
```

```
    print("Queue is empty!")
```

#Access top element from queue(Peek operation)

```
if queue:
```

```
    print("Top element of queue:",queue[0])
```

```
else:
```

```
    print("Queue is empty!")
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on win32

Type "help", "copyright", "credits" or "license()" for more information.

```
>>>
= RESTART: C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python\Queue Implementation using list.py
Queue after enqueue operation: [10, 20, 30]
Dequeued element from queue: 10
Queue after dequeue operation: [20, 30]
Top element of queue: 20
>>> |
```

Ln: 9 Col: 4

Ordered List.py - C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python\Ordered List.py (3.8.3)

File Edit Format Run Options Window Help

#Ordered list implementation

class OrderedList() :

def __init__(self): **#Simple Constructor**

self.items=[]

def insert(self,item):

i=0

while(i<len(self.items) and self.items[i]<item):

i+=1

#Valid Syntax

#i++(Invalid Syntax)

self.items.insert(i,item)

def display(self):

print("Ordered list:",self.items)

#Main function

ol=OrderedList()

ol.insert(30)

ol.insert(10)

ol.insert(20)

ol.insert(5)

#Display ordered list

ol.display()

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8

c832, May 13 2020, 22:37:02)

[MSC v.1924 64 bit (AMD64)] o

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e information.

>>>

==== RESTART: C:\Users\ADMIN\

Desktop\YMCA COLLEGE BCA\Pyth

on\Ordered List.py ===

Ordered list: [5, 10, 20, 30]

Image Various Functions.py - C:\Users\ADMIN\Desktop\YMCA COLLEGE BCA\Python\Image Various Fu

File Edit Format Run Options Window Help

```
from PIL import Image as i, ImageFilter as f
```

```
#1.Open image
```

```
img=i.open("PHOTO.jpeg")
```

```
img.show()
```

```
#2.Rotate image
```

```
rotate=img.rotate(90)
```

```
rotate.save("Images/rotate.jpg")
```

```
#3.Resize image
```

```
resize=img.resize((250,250))
```

```
resize.save("Images/resize.jpg")
```

```
#4.Convert format
```

```
img.save("Images/Photo_converted.png")
```

```
#5.Crop image
```

```
crop=img.crop((50,50,200,200))
```

```
crop.save("Images/crop.jpg")
```

```
#6.Grayscale image
```

```
gray=img.convert("L")
```

```
#'L' Grayscale Mode
```

```
gray.save("Images/grayscale.jpg")
```

```
#7.Flip image
```

```
flip=img.transpose(i.FLIP_LEFT_RIGHT)
```

```
flip.save("Images/flip.jpg")
```

```
#8.Blur image
```

```
blur=img.filter(f.BLUR)
```

```
blur.save("Images/blur.jpg")
```

```
print("Image processing completed successfully!!")
```

Add key-value pair in nested dictionary.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Add key-value pair in nested dictionary.py (3.8.3)

File Edit Format Run Options Window Help

```
#Add key-value pair in a nested dictionary
given_dict={
    "user1": {"name": "Alice", "age": 25},
    "user2": {"name": "Bob", "age": 30}
}
print(f"The given dictionary: {given_dict}")
given_dict["user3"]={"name": "Charlie", "age": 35}
updated_dict=given_dict
print(f"The updated dictionary: {updated_dict}")
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC
v.1924 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more inf
ormation.
>>>
= RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Add k
ey-value pair in nested dictionary.py
The given dictionary: {'user1': {'name': 'Alice', 'age': 25}, '
user2': {'name': 'Bob', 'age': 30}}
The updated dictionary: {'user1': {'name': 'Alice', 'age': 25},
'user2': {'name': 'Bob', 'age': 30}, 'user3': {'name': 'Charlie
', 'age': 35}}
>>> |
```



```
Tuples.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Tuples.py (3.8.3)
File Edit Format Run Options Window Help

#WAP to implement tuples in Python

#1.Create a tuple
fruits_tuple=("Apple","Banana","Cherry","Papaya")

#2.Accessing tuple element
print(f"First fruit: {fruits_tuple[0]}")
print(f"Last fruit: {fruits_tuple[-1]}")

#3.Checking the length
print(f"Number of items: {len(fruits_tuple)}")

#4.Iterating a tuple
for index,fruit in enumerate(fruits_tuple):
    print(f"{index+1}. {fruit}")

#5.Concatenating tuples
more_fruits=("Mango","Strawberry","Watermelon")
all_fruits=fruits_tuple+more_fruits
print(f"Joined tuple: {all_fruits}")

#6.Unpacking a tuple
f1,f2,f3,f4=fruits_tuple
print(f"Unpacked values: {f1}, {f2}, {f3}, {f4}")
```

```
Python 3.8.3 Shell
File Edit Shell Debug Options Window Help
Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:02) [MSC v.1924 64 bit (AMD64)] on wi
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Tuples.py =====
First fruit: Apple
Last fruit: Papaya
Number of items: 4
1. Apple
2. Banana
3. Cherry
4. Papaya
Joined tuple: ('Apple', 'Banana', 'Cherry', 'Papaya', 'Mango', 'Strawberry', 'Watermelon')
Unpacked values: Apple, Banana, Cherry, Papaya
>>> |
```

Regex_Search.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Regex_Search.py (3.8.3)

File Edit Format Run Options Window Help

```
import re
text="Science is experimental!"
pattern="Science"
search=re.search(pattern,text).group()
if search:
    print("The Pattern is found in the given text!")
    print(f"The word is {search}")
else:
    print("The Pattern is not found in the given text!")
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:0
D64)] on win32

Type "help", "copyright", "credits" or "license()" for
>>>

==== RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/P

The Pattern is found in the given text!

The word is Science

>>>


```
Regex_Replacement.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Regex_Replacement.py (3.8.3)
File Edit Format Run Options Window Help

import re
text="Dog is Cute!"
pattern="Dog"
replacement="Cat"
replace=re.sub(pattern,replacement,text)
if replace:
    print("The given text is replaced by another text!")
    print(f"Replaced text: {replace}")
else:
    print("The text isn't replaced by another text!")
```

```
Python 3.8.3 Shell
File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:37:00;
D64) on win32
Type "help", "copyright", "credits" or "license()" for more
>>>
= RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/
The given text is replaced by another text!
Replaced text: Cat is Cute!
>>> |
```

Regex_validate_email.py - C:/Users/ADMIN/Desktop/YMCA COLLEGE BCA/Python/Regex_validate_email.py (3.8

File Edit Format Run Options Window Help

```
import re
text="My ally's email is test123@gamil.com"
pattern="\w+@\w+\.\w+"
validate=re.search(pattern,text).group()
if validate:
    print("The given E-mail ID is correct!!")
    print(f"E-mail ID is: {validate}")
else:
    print("The given E-mail ID is wrong!!")
```

Python 3.8.3 Shell

File Edit Shell Debug Options Window Help

Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020
D64)] on win32

Type "help", "copyright", "credits" or "license()"

>>>

= RESTART: C:/Users/ADMIN/Desktop/YMCA COLLEGE
BCA/Python/Regex_validate_email.py

The given E-mail ID is correct!!

E-mail ID is: test123@gamil.com

>>> |